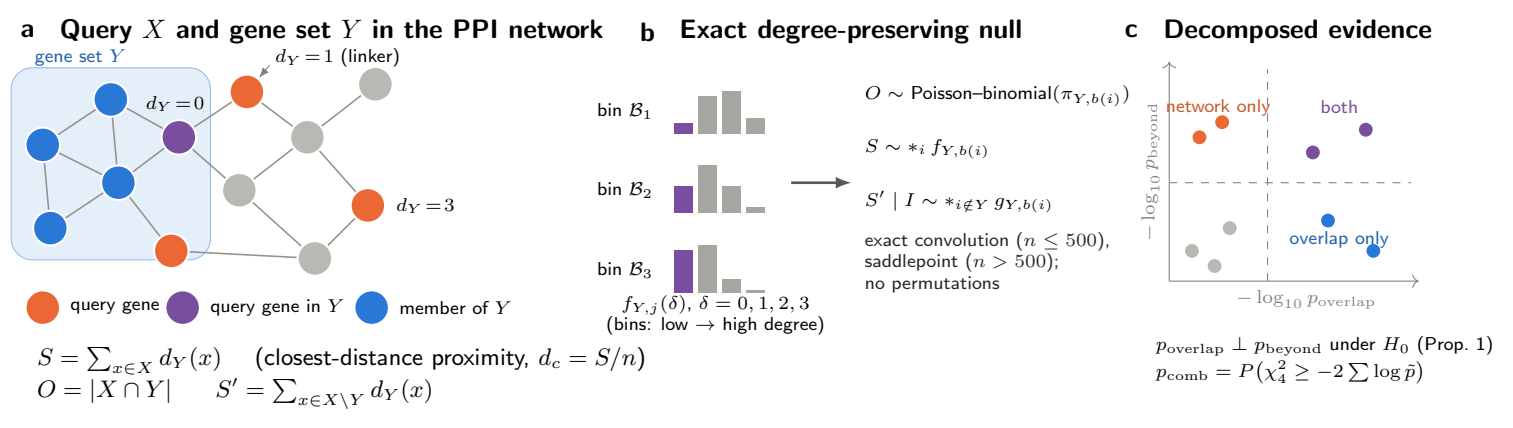

a Query X and gene set Y in the PPI network
gene set Y
$d_Y = 1$ (linker)
$d_Y = 0$
$d_Y = 3$
query gene
query gene in Y
member of Y
b Exact degree-preserving null
bin $\mathcal{B}_1$
bin $\mathcal{B}_2$
bin $\mathcal{B}_3$
$f_{Y,j}(\delta)$, $\delta = 0, 1, 2, 3$
(bins: low → high degree)
$O \sim$ Poisson–binomial$(\pi_{Y,b(i)})$
$S \sim *_i f_{Y,b(i)}$
$S' \mid I \sim *_{i \notin Y} g_{Y,b(i)}$
exact convolution ($n \leq 500$),
saddlepoint ($n > 500$);
no permutations
c Decomposed evidence
network only
both
overlap only
$-\log_{10} p_{\text{beyond}}$
$-\log_{10} p_{\text{overlap}}$
$p_{\text{overlap}} \perp p_{\text{beyond}}$ under H_0 (Prop. 1)
$p_{\text{comb}} = P(\chi^2_4 \geq -2 \sum \log \tilde{p})$
$S = \sum_{x \in X} d_Y(x)$ (closest-distance proximity, $d_c = S/n$)
$O = |X \cap Y|$ $S' = \sum_{x \in X \setminus Y} d_Y(x)$